

Impacts of Wind Energy In-Feed on Power System Small Signal Stability

Hasan Ul Banna, Alvaro Luna, Shaoqing ying, Hamidreza Ghorbani, Pedro Rodriguez
Renewable Electrical Energy Systems Research Center
Universitat Politecnica de Catalunya
Terrassa, Spain
Majorhasan_209@yahoo.com

Abstract— Integration of large amount of wind energy in an interconnected power system creates concerns about secure, reliable and economical operation of the entire power system. So it becomes very necessary to investigate the impacts of wind power infeed on the dynamic behavior of the power system. This paper presents the impacts of large amount of wind power in feed on the rotor oscillatory stability. Wind turbine generator types currently employed in wind farms, optimal location of the wind farms in the interconnected power system, reliable optimal dispatch of the wind power and the degree of tie line congestion have been thoroughly investigated. Kundur's two area network model has been utilized to study the mentioned impacts on the overall system using MATLAB/Simulink. Some of the key results show that the damping characteristics of the wind farms critically depend on the location of interconnection in the network and the optimal wind energy dispatch. Increasing the penetration of the wind energy generally improves the damping of the inter area oscillations. Moreover, reducing stress on the weak tie lines also improves the inter area mode of oscillation. So with the investigation of these impacts this study is helpful for the planning of new wind power projects.

Keywords—*Wind Farm Integration, Wind Turbine Generators, Rotor oscillatory stability, Wind Power Location,*

I. INTRODUCTION

Conventional synchronous generation dominated the power industry since too long ago but now due to the shift towards clean and more environment friendly generation, wind energy has become the topic of research, because of the absence of harmful emissions and infinite availability of prime mover. Therefore, the dynamics of the power system is drifting towards the domination of induction generators in a system with large wind energy penetration. Unlike the conventional generation, wind energy is highly intermittent and has lower inertial response as a result of unpredictable wind speeds [1]. Mostly the wind farms are usually located very far away from the load center further reducing the availability of strong grid conditions. Because the grid in these far areas are planned for unidirectional energy flow. Under these stressed and intermittent behavior of wind energy it could have a negative impact on the small signal stability of the power system.

Power oscillations are produced in a system, following a small load change or a disturbance, when there is an insufficient damping [2]. In these large interconnected system low frequency electromechanical oscillations exist due to

weak tie lines or high power transfer to long distances [3]. These oscillations may be classified in two major modes as control modes concern with the control system like exciter, governor etc and the electromechanical oscillation modes with frequency 0.1-2.0 Hz. These rotor angle modes, which are further classified as local mode (0.8-2.0Hz) and inter area mode (0.1-0.8Hz), should be damped for secure and reliable operation of the power system [4].

Wind generators don't contribute in electromechanical oscillation damping performance because of their asynchronous connection with the grid [5, 6]. But with the increase of wind power penetration it may shift the dispatch from conventional synchronous generation causing a different load flow distribution and thus can alter the damping performance of the whole interconnected system. Jones presented the scheme to control these electromechanical oscillations for HVDC links [7] and Al Biati used Flexible AC Transmission Lines (FACTS) devices to study their impacts on power oscillations [8]. L. B. Shi studied the impact of wind penetration on power system using Doubly Fed Induction Generator (DFIG) concept [9]. Ayodele discussed the effect of interconnected tie lines on oscillatory modes [10]. The influence of different generator types on power oscillations was discussed by Ghosh in [11].

In stability analysis it is necessary to define first that where the wind farm is connected in an interconnected power system, which wind generator technology is used and how strong or supportive is the grid where the wind farm is located. Different voltage and penetration levels may alter the behavior of the power system [12]. Constant speed wind turbines help to improve the oscillatory stability of the system but this contribution is limited. On the other hand, variable speed wind turbines use the control of power converters to deliver the required active and reactive power to the grid. So the use of the variable speed wind turbines is recommended to actively regulate the power in the grid [13]. Variable speed wind turbines provide this regulation by varying its output power to damp the desired oscillation mode [14].

The structure of this paper is as follows. First, the basic concepts of power system stability are described briefly. Then the different wind turbine technologies and their associated performances for system stability are discussed. In section four the structure and operating conditions of the system under consideration is presented. And then in the last section degree

of penetration of wind energy, optimal location of wind farm connection and the degree of network congestion is investigated. In the final section conclusions and discussion about the results are revised briefly.

II. POWER SYSTEM STABILITY

Power system stability is defined as the ability to remain in equilibrium during normal operating conditions and to regain an acceptable equilibrium after being subjected to a physical disturbance with most system variables bounded [15]. Broadly speaking the stability of a power system is classified as follows:

1) Voltage Stability

The capability of a power system to remain the voltages of all the buses in steady state under normal operating conditions and after a disturbance.

2) Frequency Stability

The capability of a power system to restore the balance between the system generation and the load with minimum loss of the load.

3) Rotor Angle Stability

The ability of each generating machine in an interconnected power system to maintain or regain the equilibrium between mechanical torque and the electromagnetic torque.

Rotor angle stability is further divided in two major categories depending on the severity of the disturbance. Small signal stability is usually considered for small disturbance like small change in generation or load in the power system. Two modes are observed in small signal study. Oscillatory mode which is due to lack of the damping torque and can be removed using Power System Stabilizer (PSS). And the non-oscillatory or aperiodic mode which is due to the lack of sufficient synchronizing torque and is solved using Automatic Voltage Regulator (AVR) [16].

III. WIND TURBINE GENERATOR TECHNOLOGY

The basic working principle of a wind turbine involves two conversion processes. The turbine's rotor converts the kinetic energy of wind into rotating movement which is further converted to electricity by a generator.

A. Squirrel Cage Induction Generator

In this type of constant speed wind turbine technology turbine's rotor is coupled with the generator through a gear box and the stator is directly connected to the grid as shown in fig. 1. The power extracted from the wind is limited due to stall effect of the rotor's blades angle control. And as SCIG has high reactive power consumption so it is usually equipped with capacitor banks further maintain the voltage at the grid. The rotor speed of SCIG varies with the output power but these variations in rotor speed are usually very small and that's why SCIG is considered as constant speed turbine generator. However, by changing the poles of the stator it can be run at two different speeds [17].

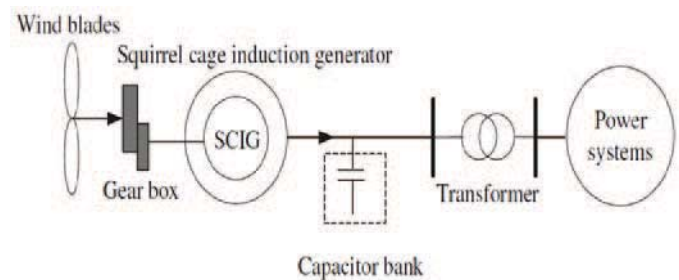


Fig. 1. Schematic Diagram of Squirrel Cage Induction Generator

B. Doubly Fed Induction Generator

It is a variable speed pitch control turbine consisting of a wind turbine, drive train, gear box, induction generator and back to back voltage source Pulse Width Modulation (PWM) converter. The rotor is connected to the grid through PWM converter and the stator is directly connected to the grid as shown in the fig. 2. The active and reactive power of DFIG is controlled through Rotor Side Converter (RSC) and Grid Side Converter (GSC) which further improves the wind energy conversion efficiency and the voltage stability as well [18]. But due to small internal resistance of DFIG the capability of oscillation damping is limited.

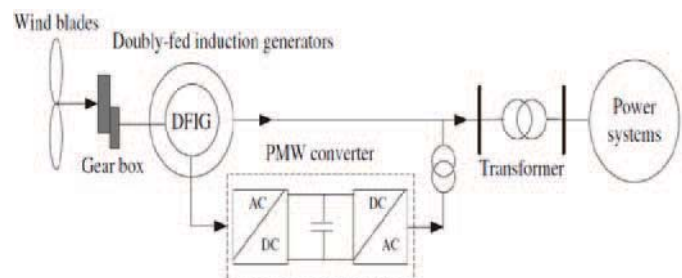


Fig. 2. Schematic Diagram of Doubly Fed Induction Generator

C. Directly Driven Synchronous Generator

This type of variable speed wind turbine consists of wind synchronous generator, drive train and full scale back to back PWM converter. The synchronous generator may have a wound rotor or be excited using permanent magnet. The rotor is directly connected to the generator and the stator is coupled with the grid through PWM converter as shown in the fig. 3. The simple structure of DDSG reduces the maintenance cost and provides improved operational reliability. The output power of DDSG is controlled through pitch angle control and PWM full scale converter which makes it electrically decoupled from the grid [19].

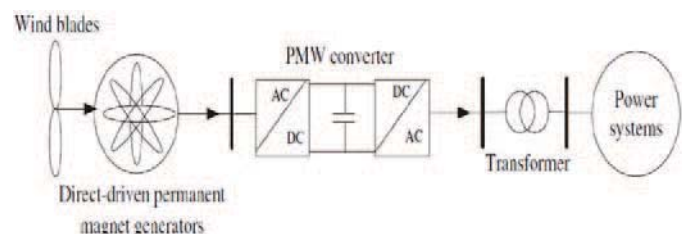


Fig. 3. Schematic Diagram of Directly Driven Synchronous Generator

A performance comparison between these wind turbine technologies is shown in the table. 1

Table 1. Performance of WTG Technologies

	Advantages	Disadvantages
SCIG	Low cost, Simple and robust construction, Electrically efficient	Inductive power demand, Noisy, High Mechanical stress, Aerodynamically less efficient, Losses in Gear box
DFIG	Aerodynamically high energy yield, Small rating converter, Less noisy	High maintenance cost, Electrically less efficient, Losses in Gear box
DDSG	Less noisy, Aerodynamically efficient, No Gear box losses, Less mechanical stress	High cost for permanent excitation, Large rating converter needed, Heavy and large generator

IV. SYSTEM UNDER STUDY

The system under consideration is shown in fig. 4. It has been taken from [20] which is a simple yet realistic interconnected model specially designed for power oscillations studies [21]. It consists of 11 buses and two symmetrical areas which are connected through two weak tie lines. Two shunt capacitors and two loads are connected at each of the buses 7 and 9. Each area has two sets of synchronous generators. G1, G2 in Area 1 and G3, G4 in Area 2. Each synchronous generator induces 900MVA with 20KV rating and is represented by the same dynamic model. Active power transfer of 413MW is being considered from Area 1 to Area 2. Generators' output for G1, G2, G3 is 700MW and 719MW for G4. Area1 and Area2 loads are 967MW and 1767MW respectively. The system has same dynamic data of generators as in [20]. For Simulation and analysis purpose

only DFIG turbine technology is used in this research. The turbine model and all parameters have been taken from [22].

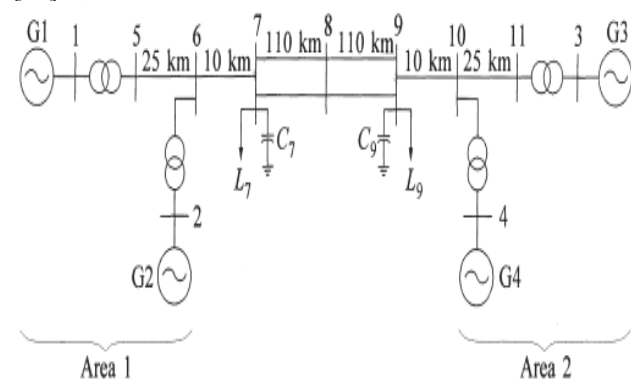


Fig.4 Two Area Power System

V. RESULTS AND DISCUSSIONS

Electrical power of each synchronous generator has been plotted against time for different scenarios of network configurations to study the impacts of wind energy on damping of inter area oscillations.

A. Penetration Level of Wind Energy

1) Scenario 1 (Base Case)

Base case is when no wind farm is integrated in the network and all of the synchronous generators are operating with their full output power as described in the section four. There is an inter area flow of power from Area 1 to Area 2 due to the difference in supply and load in each area. So an amount of 413MW power flows from Area 1 to Area 2.

Because of the inherent nature of Kundur's model there are no inter machine oscillations between G3 and G4, as oscillations in electrical power output of both the generators are oscillating in the same directions with the same frequency as shown in fig.5. But for G1 and G2 there are initially some oscillations in their electrical power output, but these oscillations are damped till $t=4$ sec. From $t=4$ sec we can see that there are noticeable inter area oscillations between Area 1 (G1 and G2) and Area 2 of increasing magnitude. Both the group of generators G1, G2 and G3, G4 are oscillating in opposite direction as shown in fig.5.

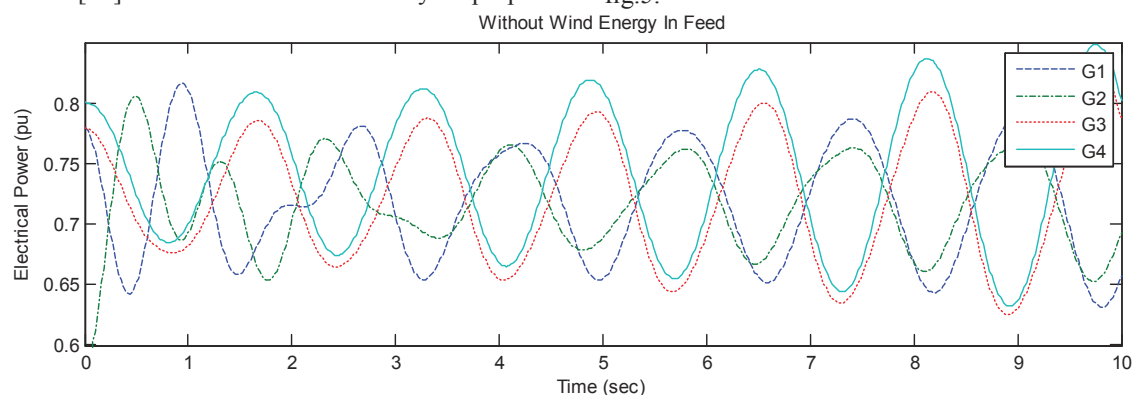


Fig. 5 Electrical Power Output of Synchronous Generator

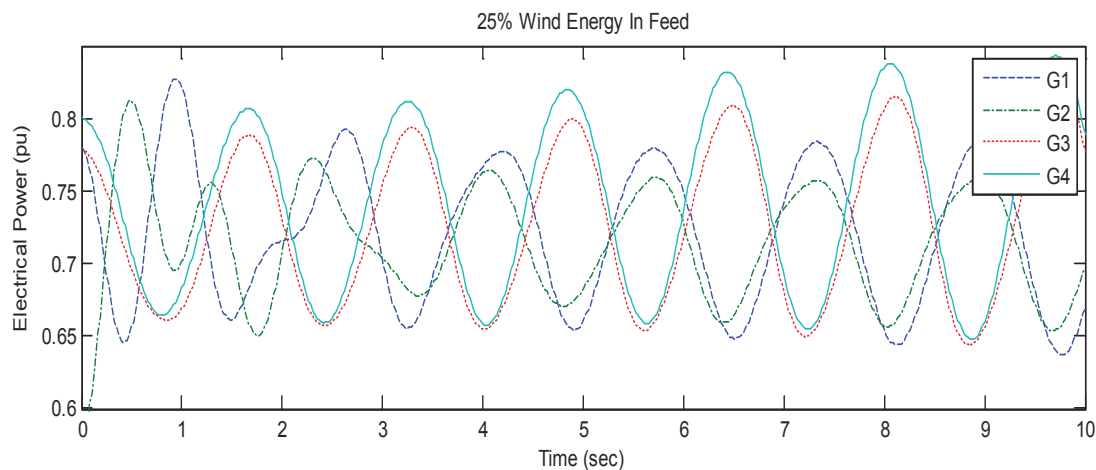


Fig. 6. Electrical Power Output of Synchronous Generator

2) Scenario 2 (25% Wind Energy)

In this case 25% power from conventional synchronous generators is replaced with wind energy by connecting DFIG turbines at each of the buses 5,6,10 and 11. G1, G2 and G4 were producing 700MW active power in the previous scenario and G3 was producing 719MW. Now after integrating wind power at above mentioned buses these synchronous generators will be delivering 75% power as compare to the previous case and the rest of the 25% power for each synchronous generator will be fed through DFIG wind turbines.

We can see in the fig. 6 that wind energy itself doesn't induce any of the oscillations mode rather it helps in damping the existing oscillations as the overall synchronous behavior is replaced with induction generators, which are not directly coupled with the grid rather connected through back to back converter.

3) Scenario 3 (50% Wind Energy)

As described the procedure for integrating the wind energy at buses 5,6,10 and 11, here in this case the amount of wind energy in feed has increased to 50% of the total power of each synchronous generator in the base case. The rest of the 50% power is still be delivering through synchronous generators. From the fig. 7 it is evident that the damping of inter area oscillations (between G1, G2 and G3, G4) has increased and there is a slight increase in the frequency as well. It is because the concentration of DFIG turbine is increasing and DFIG based wind turbines are partially decoupled from the grid through its rotor [24]. So the magnitude of oscillations in electrical power output is decreasing.

4) Scenario 4 (75% Wind Energy)

In this case the amount of wind energy in-feed has increased to 75% of the total power of each synchronous generator in the base case. The rest of the 25% power is still be delivering through synchronous generators.

The damping of inter area oscillations (between G1, G2 and G3, G4) has increased considerably and there is an increase in the frequency too.

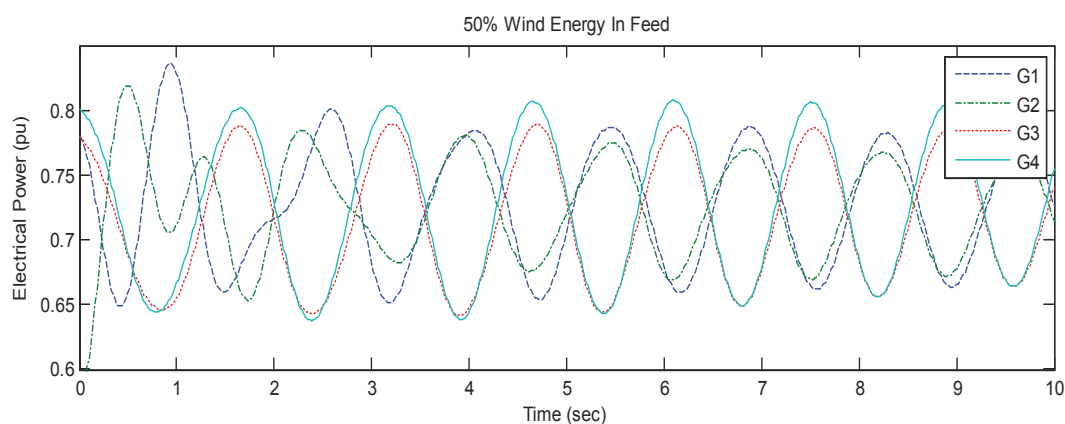


Fig. 7. Electrical Power Output of Synchronous Generator

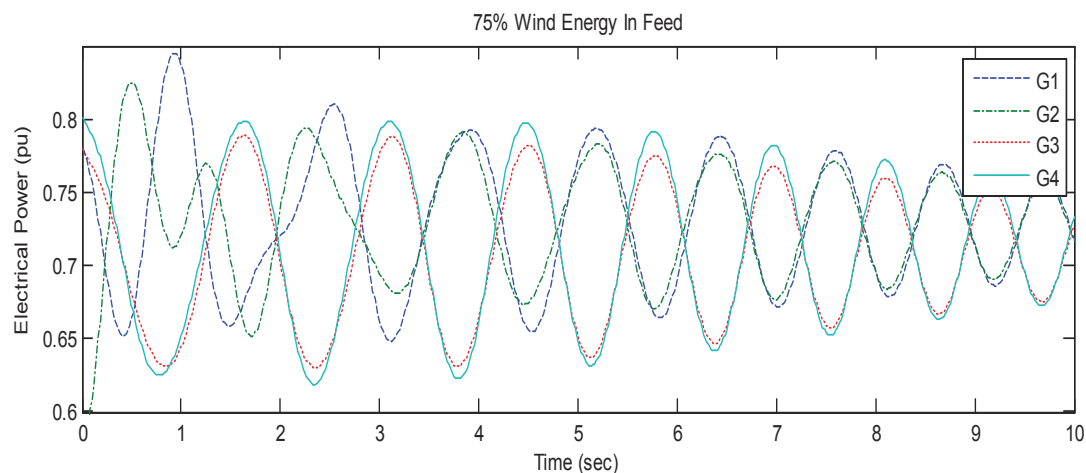


Fig. 8. Electrical Power Output of Synchronous Generator

As the wind energy in feed has reached to 75% of the overall power production and now it's dominating in the whole network. The magnitude of oscillations in electrical power is decreasing rapidly as shown in the fig.8.

B. Location of Wind Farm Infeed

1) Wind Farm in an Area Exporting Power

Area 1 delivers 413MW of active power to Area 2, because of the difference in the production and load in each area. To study the impact of wind farm location on the power network, wind farm is first connected at bus 7 which acts as the boundary of Area 1, which is exporting power to Area 2. In fig.9 we can see that by installing wind energy farm in Area 1, the local mode of oscillations which were initially present between G1 and G2 are completely vanished. This is due to the fact that DFIG doesn't induce any additional mode of oscillation because it is partially decoupled from the grid [25]. But the inter area oscillations are still present in the network and the magnitude of inter area oscillations has increased as

well because it involves the oscillations between distant areas (Area1 and Area2) and by installing wind turbines just in one area, the oscillations between them can't be damped properly. So by installing wind farm in an area which is exporting power to some other nearby area, inter machine oscillations are damped.

2) Wind farm in an Area Importing Power

For this case wind farm is connected at bus 9 acting as the starting point of Area 2, which is importing power from Area 1. When the wind farm is connected in the area which is importing the power from other nearby area then it has very little effect on the inter machine oscillations of the other area. But the frequency of inter area oscillations increases in comparison with the connection when it was connected in an area which was exporting the power as shown in the fig.10. However, the damping doesn't change considerably.

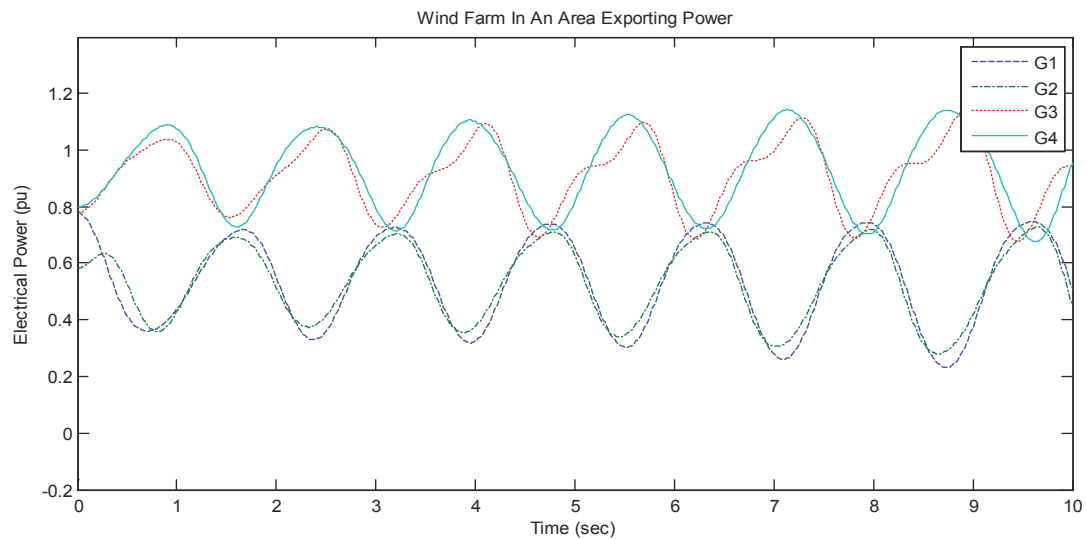


Fig. 9. Electrical Power Output of Synchronous Generator

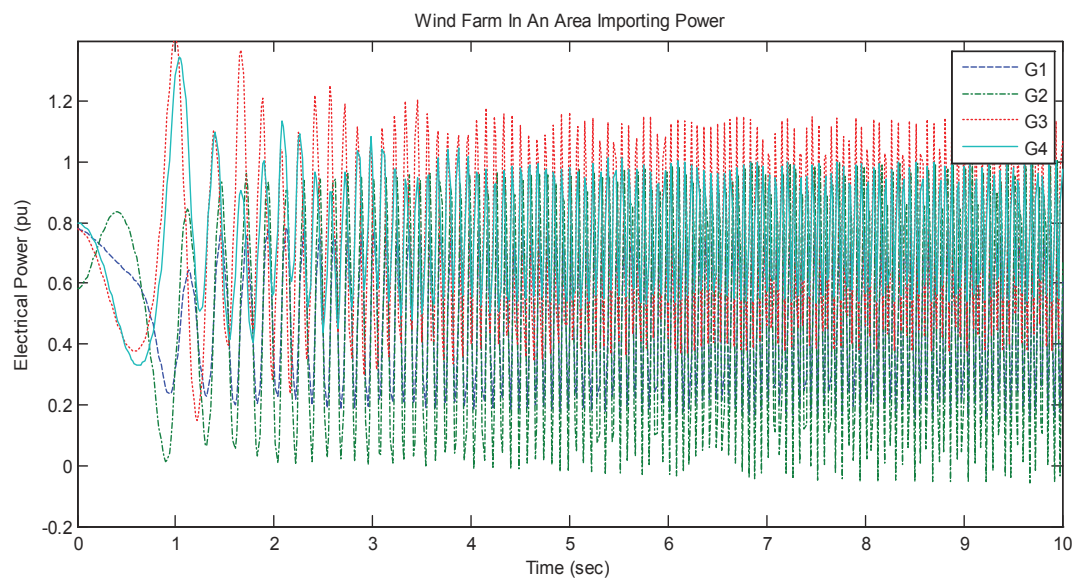


Fig. 10. Electrical Power Output of Synchronous Generator

C. Stress On The Tie Line

1) Relaxed Tie Line

In the considered power network there are two tie lines connecting Area 1 and Area 2 each of 220km long. There are 413MW flowing in the lines and the associated oscillations are shown in the fig.11

2) Stressed Tie Line

If one of the tie lines connecting the two areas is removed the stress level on the remaining single tie line will be increased. It doesn't have any effect on the damping but the magnitude and the frequency of inter area oscillations decrease considerably as shown in the fig.12. So the degree of congestion of the tie lines connecting the areas is very critical for inter area mode of oscillations as inter area mode determines the stability of the entire system [26].

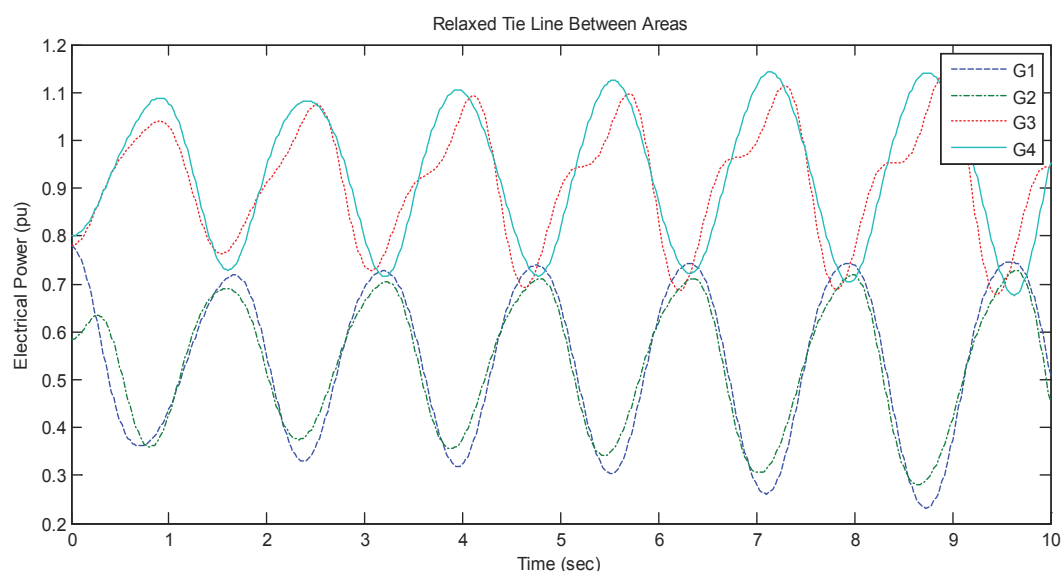


Fig. 11. Electrical Power Output of Synchronous Generator

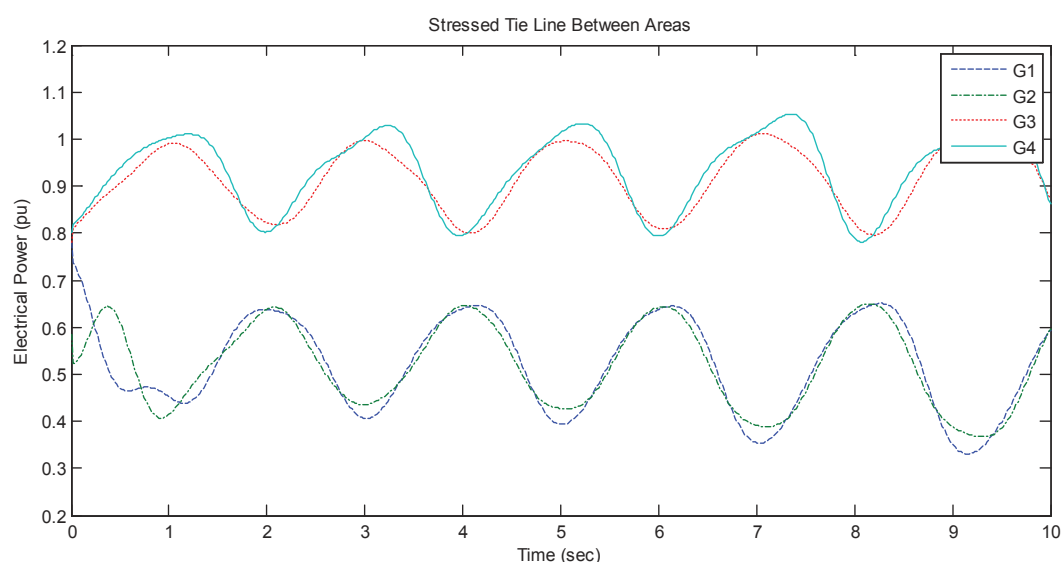


Fig. 12. Electrical Power Output of Synchronous Generator

VI. CONCLUSION

In this paper the impacts of intermittent wind energy on power system oscillation stability has been investigated. The impacts of wind energy in an interconnected network can be positive or negative depending upon the penetration level, location and the degree of stressed on the tie line connecting the areas. The power system model used for investigation, has intrinsic oscillations due to specific load distribution and stress level of the tie lines.

Increased level of wind energy farms based on DFIG increases the damping performance of the inter area oscillations. Moreover, DFIG doesn't induce any additional oscillation modes in the network as it is partially decoupled from the grid

through its rotor. With increased penetration of DFIG based wind farms the effective overall inertia of the network is reduced as DFIG are not fully coupled with the grid as synchronous generator are. So the network's reliability following a disturbance may be significantly affected. When the synchronous generators in any specific area are fully replaced with DFIG based wind farm the local mode oscillations of that area are fully damped. By enhancing the damping of inter area oscillations wind energy integration helps in reducing the stress level on the tie lines. Wind farm in an area exporting power not only improves the local oscillations of that area but also decreases the frequency of inter area oscillations. However, it doesn't affect the local mode of other adjacent. The degree of congestion of the tie lines connecting the areas is very critical for inter area mode

as this mode of oscillations determines the stability of the entire system.

ACKNOWLEDGMENT

This project has been partially supported by the Spanish Ministry of Science under the project ENE2011-29041-02-01.

REFERENCES

- [1] G. Papaefthymiou et al. , "Integration of Stochastic Generation in power Systems", International Journal OF Electric Power and Energy Systems, vol. 28, pp. 655-667, 2006
- [2] M. Klein, G. J. Rogers, P. Kundur, "A Fundamental Study of Inter area Oscillations in Power Systems" IEEE Transactions on Power Systems, vol. 6, No. 3, August 1991.
- [3] A. Mendonca and J.A.P. Lopes, "Impact of Large Scale Wind Power Integration on Small Signal Stability", International Conference on Future Power Systems, Amsterdam, 2005, pp.5
- [4] P. Kundur, M. Klein, G.J. Roger, M.S. Zywno, "Application of Power System Stabilizers For Enhancement Of Overall System Stability", IEEE Transactions on Power Systems, vol. 4, No. 2, May 1989
- [5] J. G. Slooetweg and W.L. Kling, "The Impact of Wind Power Generation on Power System Oscillations", Electric Power System Research, vol.67, pp. 9-20, 2003
- [6] J. Stenzel et al., "Impact of large Scale Wind Power on Power System Stability", 5th International Workshop on Large-Scale Integration of Wind and Transmission Network for Offshore Wind Farm, Glasgow, Scotland 2005
- [7] L. E. Jones , "On Zero Dynamics and Robust Control of Large AC and DC Power Systems", PhD Thesis, Royal Institute of Technology, Stockholm, 1999
- [8] M.A. Al-Biati, M.A. El-Kady, A.A. Al-Ohal, "Dynamic Stability Improvement via coordination of SVC and PSS Control Actions", Electric Power System Research, vol. 58, No. 1, pp 37-44, 2001
- [9] L. B. Shi. Et al. "Analysis of Impacts of Grid Connected Wind Power on Small Signal Stability", Wind Energy 2010
- [10] T.R. Ayodele, A.A. Jimoh, J.L. Munda, J. T. Agee, "Impacts of Tie Lines and Wind Generator Location on Small Signal Stability of a Power System", International Journal of Renewable Energy Research, vol.3, No. 1, 2013
- [11] S. Ghosh and N. Senroy, "Effect of Generator Type on the participation of Adjacent Synchronous Machines in Electromechanical Oscillations", Joint International Conference on Power Electronics, Drive and Energy Systems (PEDES) 2010
- [12] Thakur D. and Mithulanaathan, "Influence of Constant Speed Wind Turbine Generator on power System Oscillations", Electric Power Components and Systems, vol. 37, No. 5, pp 478-494
- [13] Martinez, C. Joos G. and Ooi B.T. "Power System Stabilizers in Variable Speed Wind Farms", Power and Energy Society General Meeting, Calgary 2009
- [14] Miao, Z. Fan, L. Osborn and Yuvarajan S, "Control of DFIG-based Wind Generation to Improve Inter area Oscillation Damping", IEEE Transaction on Energy Conversion, vol. 24, No. 2, pp 415-422
- [15] Kundur P., Paserba J. and Vitet S., "Overview on Definition and Classification of Power System Stability", Proceedings of Quality and Security of Electric Power Delivery Systems CIGRE/IEEE PES International Symposium 2003
- [16] IEEE/CIGRE Joint Task Force on Stability Terms and Definitions, "Definition and Classification of Power System Stability", IEEE Transactions on power Systems, vol. 19, No.2, pp 1387-1388
- [17] Joselin Herbert GM, Iniyar S, Sreevalsan E, "A review of Wind Energy Technologies", Syst Energy Rev 11, No. 6, pp 1117-1145
- [18] Yang L.H., Xu Z, Ostergaard J, "Advanced Control Strategy of DFIG Wind Turbines for Power System Fault Ride Through", IEEE Trans Power System, vol. 27, No. 2, pp 713-722
- [19] J.G. Slooetweg, W.L. Kling, "The Impact of Large Scale Wind Power Generation on Power System Oscillations", Electric Power System Research, vol. 67, pp 09-20 2003
- [20] Kundur P. (1994). Power System Stability and Control. McGraw-Hill, pp : 710-830
- [21] M. Klein, G.J. Roger, S. Moorthy, P. Kundur, "Analytical Investigation of Factors Influencing Power System Stabilizers performance", IEEE Transactions on Energy Conversion, vol. 7, No. 3, September 1992
- [22] N. W. Miller, J. J. Sanchez-Gasca, W. W. price, R. W. Delmerico, "Dynamic Modeling of GE 1.5 and 3.6 MW Wind Turbine-Generators for Stability Simulations", proc. Power Engineering Society General Meeting, Toronto Ontario, July 2003
- [23] Espen Hagstrom, Ian Norheim and Kjetil Uhlen, "Large Scale Wind Power Integration in Norway and Impact on Damping in the Nordic Grid", Wind Energy, Iss. 8, pp. 375-384, 2005
- [24] Marzio Di, olav B., Kjetil Uhlen, Magni Palsson, "Large Scale Wind power Integration, Voltage Stability limits and Modal Analysis", 15th PSCC, Liege Belgium, 22-26 August 2005
- [25] Jose L. Rueda, Istvan Erlich, "Impacts of Large Scale Integration of Wind Power on Power System Small Signal Stability", 4th International Conference on Electric Utility Deregulation and Restructuring and Power Technologies, July 2011, Weihai, Shandong, China